

# Ibrahim Kecoglu

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🌐 ikecoglu | 🌐 ibrahimkecoglu |

## EDUCATION

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PhD, Biomedical Physics at **Stanford University** 2024 - 2030  
BS, Mathematics at **Bogazici University** (High Honors, GPA: 3.79/4.00) 2022 - 2024  
BS, Physics at **Bogazici University** (High Honors, 2<sup>nd</sup> Place, GPA: 3.79/4.00) 2020 - 2024

## RESEARCH EXPERIENCE

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**Durmus Lab – Stanford University (Supervisor: Gozde Durmus)**  
**Graduate Researcher**

July 2025 – Present

- Built a magnetic levitation Raman spectroscopy platform to sort and characterize microliter-scale biological samples.

**PhD Lab Rotation**

Sep. 2024 – Dec. 2024

- Developed computational simulations to study cell aggregation in oscillating-field magnetic levitation devices.
- Optimized SERS substrate performance by evaluating surface treatment protocols.
- Applied SERS for exosome detection, improving the sensitivity of nanoscale vesicle analysis.
- Gained hands-on wet lab experience, including cell culture, biosafety cabinet practices, and quantitative cell assays.

**Loo Lab – Stanford University (Supervisor: Billy W. Loo)**  
**PhD Lab Rotation**

March 2025 - June 2025

- Designed and constructed a lateral beam irradiation setup for mouse brain studies.

**Molecular Imaging Instrumentation Laboratory – Stanford University (Supervisor: Craig S. Levin)**  
**PhD Lab Rotation**

Jan. 2025 – Mar. 2025

- Developed a generative deep learning model to synthetically enhance PET images, producing higher-resolution outputs from conventional PET data.

**Berker Research Group – MIT & Kadir Has University (Supervisor: Ahmet Nihat Berker)**  
**Undergraduate Researcher**

Mar. 2020 – June 2024

- Investigated statistical mechanics models such as Ising, Potts, and Ashkin–Teller.
- Characterized phase diagrams and transitions.
- Applied advanced methods such as renormalization group theory, duality transformations, transfer matrix methods, Migdal–Kadanoff approximation, and multifractal analysis.
- Developed a novel framework for classifying chaotic behavior in real-world systems.
- Authored three peer-reviewed publications (two first-author, one co-author).

## Medical and Biological Physics Laboratory – Bogazici University (Supervisor: Mehmet Burcin Unlu)

Undergraduate Researcher

Aug. 2019 – June 2024

- Designed and built scanning Raman spectroscopy, optical tweezer, and microscopy setups.
- Conducted experiments on diverse biological samples, including animal tissues, blood, plant tissues, and nano-scale vesicles (exosomes).
- Applied spectral analysis methods (e.g., band component analysis) and built statistical and machine learning models in MATLAB and Python.
- Performed biological sample preparation, including extracellular vesicle isolation from cell media.
- Authored seven peer-reviewed publications (one first-author, six co-author).

## TEACHING EXPERIENCE

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Stanford University - Teaching Assistant

Sep. 2025 - Dec. 2025

- BMP/RAD/BIOE 222: Physics and Engineering Principles of Multi-modality Molecular Imaging of Living Subjects

KHAS University Summer/Winter School - Teaching Assistant

July 2019 - June 2024

- Phase Transitions and Renormalization-Group Theory
- Electricity and Magnetism
- Introduction to Quantum Mechanics
- Introduction to Computer Engineering

## AWARDS & SCHOLARSHIPS

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TUBITAK 2205 - Undergraduate Scholarship Program

Sep. 2019 - June 2024

Awarded to students in the top 1% nationwide pursuing STEM studies.

EpiSTEM Research Competition - 1<sup>st</sup> Place

Apr. 2019

Designed and built a DIY spectrometer and investigated the effect of the storage surface on the decomposition of fruits.

## LICENSES & CERTIFICATIONS

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| Introduction to Molecular Biology and Genetics Engineering at <b>Kadir Has University</b>       | 2020 |
| Social and Physical Networks, Complex Systems, Chaos and Control at <b>Kadir Has University</b> | 2020 |
| Phase Transitions and Renormalization Group at <b>Kadir Has University</b>                      | 2020 |
| Big Data - Intelligent Data: You Too Can Create It, Manage It at <b>Kadir Has University</b>    | 2019 |
| Introduction to Computer Engineering at <b>Kadir Has University</b>                             | 2019 |
| Introduction to Quantum Mechanics at <b>Kadir Has University</b>                                | 2018 |

# SKILLS

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## Technical Skills

- **Programming & Data Analysis:** Python (NumPy, PyTorch, scikit-learn, SymPy), MATLAB, COMSOL, Mathematica, spectral data analysis, instrument control
- **Machine Learning & AI:** CNNs, U-Net, GANs, diffusion models, Random Forests, dimensionality reduction (PCA, t-SNE)
- **Experimental Techniques:** Photonics, Raman spectroscopy, optical tweezers, magnetic levitation, PET/CT imaging
- **Laboratory Skills:** Biological sample preparation, extracellular vesicle isolation, cell culture, SERS substrate preparation, microscopy, biosafety cabinet work
- **Theoretical & Computational Physics:** Statistical mechanics, renormalization-group theory, duality transformations, transfer matrix methods, multifractal analysis, spin-glass theory

## Language Skills

- Turkish: Native
- English: Advanced academic proficiency (10 peer-reviewed publications in English)

# PUBLICATIONS

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- [10] Arda Inanc, Nayce Ilayda Bektas, **Ibrahim Kecoglu**, Ugur Parlatan, Begum Durkut, Melike Ucak, Mehmet Burcin Unlu, and Ciler Celik-Ozenci. “Label-free differentiation of functional zones in mature mouse placenta using micro-Raman imaging”. In: *Biomedical Optics Express* 15.5 (Apr. 2024), pp. 3441–3456. ISSN: 2156-7085. DOI: [10.1364/BOE.521500](https://doi.org/10.1364/BOE.521500).
- [9] **Ibrahim Kecoglu** and A. Nihat Berker. “Global Ashkin–Teller phase diagrams in two and three dimensions: Multicritical bifurcation versus double tricriticality—endpoint”. In: *Physica A: Statistical Mechanics and its Applications* 630 (Nov. 2023), p. 129248. ISSN: 0378-4371. DOI: [10.1016/j.physa.2023.129248](https://doi.org/10.1016/j.physa.2023.129248).
- [8] E. Can Artun, **Ibrahim Kecoglu**, Alpar Türkoğlu, and A. Nihat Berker. “Multifractal spin-glass chaos projection and interrelation of multicultural music and brain signals”. In: *Chaos, Solitons & Fractals* 167 (Feb. 2023), p. 113005. ISSN: 0960-0779. DOI: [10.1016/j.chaos.2022.113005](https://doi.org/10.1016/j.chaos.2022.113005).
- [7] Ayse SEN, **Ibrahim Kecoglu**, Muhammad Ahmed, Ugur Parlatan, and Mehmet Burcin Unlu. “Differentiation of advanced generation mutant wheat lines: Conventional techniques versus Raman spectroscopy”. In: *Frontiers in Plant Science* 14 (Feb. 2023). ISSN: 1664-462X. DOI: [10.3389/fpls.2023.1116876](https://doi.org/10.3389/fpls.2023.1116876).
- [6] Ugur Parlatan, Mehmet Ozgun Ozen, **Ibrahim Kecoglu**, Batuhan Koyuncu, Hulya Torun, Davod Khalafkhany, Irem Loc, Mehmet Giray Ogut, Fatih Inci, Demir Akin, Ihsan Solaroglu, Nesrin Ozoren, Mehmet Burcin Unlu, and Utkan Demirci. “Label-Free Identification of Exosomes using Raman Spectroscopy and Machine Learning”. In: *Small* 19.9 (Jan. 2023), p. 2205519. ISSN: 1613-6829. DOI: [10.1002/smll.202205519](https://doi.org/10.1002/smll.202205519).
- [5] Irem Loc, **Ibrahim Kecoglu**, Mehmet Burcin Unlu, and Ugur Parlatan. “Denoising Raman spectra using fully convolutional encoder–decoder network”. In: *Journal of Raman Spectroscopy* 53.8 (May 2022), pp. 1445–1452. ISSN: 0377-0486, 1097-4555. DOI: [10.1002/jrs.6379](https://doi.org/10.1002/jrs.6379).

- [4] **Ibrahim Kecoglu**, Merve Sirkeci, Mehmet Burcin Unlu, Ayse Sen, Ugur Parlatan, and Feyza Guzelcimen. “Quantification of salt stress in wheat leaves by Raman spectroscopy and machine learning”. In: *Scientific Reports* 12.1 (May 2022), p. 7197. ISSN: 2045-2322. DOI: [10.1038/s41598-022-10767-y](https://doi.org/10.1038/s41598-022-10767-y).
- [3] Ugur Parlatan, Seyma Parlatan, Kubra Sen, **Ibrahim Kecoglu**, Mustafa Ozer Ulukan, Atalay Karakaya, Korhan Erkanli, Halil Turkoglu, Murat Ugurlucan, Mehmet Burcin Unlu, and Bukem Tanoren. “Atrial fibrillation designation with micro-Raman spectroscopy and scanning acoustic microscope”. In: *Scientific Reports* 12.1 (Apr. 2022), p. 6461. ISSN: 2045-2322. DOI: [10.1038/s41598-022-10380-z](https://doi.org/10.1038/s41598-022-10380-z).
- [2] Bukem Tanoren, Ugur Parlatan, Melita Parlak, **Ibrahim Kecoglu**, Mehmet Burcin Unlu, Didem Melis Oztas, Mustafa Ozer Ulukan, Korhan Erkanli, and Murat Ugurlucan. “Aortic aneurysm evaluation by scanning acoustic microscopy and Raman spectroscopy”. In: *Analytical Methods* 13.39 (Sept. 2021), pp. 4683–4690. ISSN: 1759-9660, 1759-9679. DOI: [10.1039/D1AY01133B](https://doi.org/10.1039/D1AY01133B).
- [1] **Ibrahim Kecoglu** and A. Nihat Berker. “Across dimensions: Two- and three-dimensional phase transitions from the iterative renormalization-group theory of chains”. In: *Physical Review E* 102.3 (Sept. 2020), p. 032134. ISSN: 2470-0045, 2470-0053. DOI: [10.1103/PhysRevE.102.032134](https://doi.org/10.1103/PhysRevE.102.032134).